



2025

F22

BOOKLET NO.

130032

**Animal Husbandry and
Veterinary Science
Paper – I**

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

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**SECTION – A**

- Q1.** Answer the following questions in about **150** words **each**. **10×5=50**
- Q1.** (a) Describe the walk-along (Sliding filament) theory of muscle contraction. **10**
- Q1.** (b) Describe the mechanism of Respiration in mammals. **10**
- Q1.** (c) Explain hemopoiesis. Write stages, regulation and physiological significance of erythropoiesis in mammals. **10**
- Q1.** (d) Define oestrus. Describe its causes, types and management in farm animals with examples. **10**
- Q1.** (e) Explain the importance of fodder in animal production. Briefly describe the sequence of agronomical practices involved in the cultivation of fodder crops. **10**
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- Q2.** (a) Define renal failure. Describe its types, causes, pathophysiology, clinical manifestation and biochemical test used for diagnosis. **20**
- Q2.** (b) Discuss how modernization has simplified livestock farm management viz a viz improved productivity and increased profitability. **20**
- Q2.** (c) Describe the blood circulation in birds with special reference to structural adaptations of the heart and blood vessels that support high metabolic demand. **10**
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- Q3.** (a) Explain the physiological basis of growth in livestock. Discuss prenatal and postnatal stages, growth curve, measurement techniques and factors affecting it. **20**
- Q3.** (b) Discuss the principles and procedures of artificial insemination in farm animals. Emphasize the role of artificial insemination in genetic improvement of livestock. **20**
- Q3.** (c) Describe commercial egg production with reference to feeding, management and economics of layer farming. **10**
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- Q4. (a) Explain the interrelationship between endocrine system and reproduction in animals. 20
- Q4. (b) Describe the mechanisms of thermoregulation in domestic animals. Discuss the effects of environmental stress on health, reproduction and productivity. 20
- Q4. (c) Discuss the role of livestock in farmer's economy and major challenges faced by dairy farming in India. 10
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SECTION – B

- Q5. Answer the following questions in about 150 words each. 10×5=50
- Q5. (a) Write about feeding of swine for lean meat production. 10
- Q5. (b) Differentiate between inheritance of qualitative and quantitative traits with suitable example. 10
- Q5. (c) Enlist the objectives of extension education. Discuss the principles of extension education and their relevance in animal husbandry extension. 10
- Q5. (d) Describe direct and indirect calorimetry used for measurement of energy metabolism for maintenance. 10
- Q5. (e) Elucidate Mendel's Law of Independent Assortment using suitable example of dihybrid cross with Punnett checkerboard. 10
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- Q6. (a) Discuss the use of Non-Protein Nitrogen (NPN) compounds in ruminant diets. Describe their sources, metabolism, advantages, limitations and precautions during feeding. 20
- Q6. (b) Define heritability (h^2) and distinguish between broad sense and narrow sense heritability. Discuss the methods of h^2 estimations, its uses and common misconceptions about heritability. 20
- Q6. (c) What are feed additives ? Classify with examples. Describe feed additives that enhance growth and improve performance in animals. 10
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- Q7.** (a) Discuss recent advances in feed technology and feed processing. Explain how these advancements improved feed utilization and livestock productivity. **20**
- Q7.** (b) Classify the systems of mating used in animal breeding with examples. Discuss the genetic and phenotypic consequences of inbreeding. **20**
- Q7.** (c) Discuss the problems and constraints in transfer of animal husbandry technologies in rural areas and suggest suitable remedial measures. **10**
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- Q8.** (a) Give the feeding strategies for milking cows during different stages of the lactation cycle. Explain the effect of nutrition on milk yield and milk composition. **20**
- Q8.** (b) What is mutation ? Classify the different types of mutations and explain methods used to detect mutations. **20**
- Q8.** (c) Classify the minerals with example. Discuss in detail about role of calcium and phosphorus in animal health and production. **10**
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